

Cycle Care

Member 3 Complete Guide

Private Education Subsystem

Offline • Private • Bilingual (English & Sinhala)

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SLIIT IT3060 Human Computer Interaction – Assignment 2, Milestone II

Branch: `feature/member3-private-education`

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1. Project & subsystem overview

Cycle Care is an offline-capable menstrual health app for teenage girls in rural Sri Lankan schools, many using a *shared family phone* with poor connectivity. Five principles apply to every screen: **offline-first, privacy by design, low cognitive load, bilingual, calm non-stigmatising visuals**.

#	Subsystem	Owner
1	Privacy & Onboarding (language, PIN setup, profile)	C G Gunasekera
2	Core Tracking (calendar, daily check-in, dashboard)	W V D S U Helanjith
3	Private Education (PIN entry, Education Hub, articles, Offline Library, Myth Buster)	A M S P Aththanayake
4	Healthcare Access (pharmacy directory)	K G L S Senadeera

Requirements covered: **REQ-3.1** (secure, discreet entry on shared phones), **REQ-3.2** (vital health information available offline), plus FR6 (offline), FR8 (Sinhala + English) and FR9 (PIN protection).

2. What Member 3 built

Screen	Class	What it does
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Privacy Lock	<code>PrivacyLockScreen</code>	4-digit PIN pad, auto-verify on 4th digit, shake on error, 5 wrong attempts → 30 s lockout (persisted).
Education Hub (Learn tab)	<code>EducationHubScreen</code>	Greeting, offline search, date-deterministic Daily Health Tip, 2×2 Explore Topics grid with live saved count.
Category list	<code>CategoryScreen</code>	Data-driven filter chips and article rows for <code>food-diet</code> and <code>our-body</code> .
Article reader	<code>ArticleScreen</code>	Hero, two-toned title, rich blocks, Village Tip callout, reversible “Save To Offline Library”, toast, panic HIDE.
Offline Library	<code>OfflineLibraryScreen</code>	“Zero Internet Required” banner, saved list (newest first), Read Offline pill, un-save, empty state.
Myth Buster	<code>MythBusterScreen</code>	5-question shuffled quiz, MYTH/FACT stacked buttons, feedback card, results, personal best.
Test overlay	<code>TestModeOverlay</code>	Floating metrics badge for user testing (only with <code>TEST_MODE=true</code>).

Shared widgets in `lib/widgets/`: `PillButton`, `AppChip`, `AppBar`, `HideButton`, `TopicTile`, `ArticleRow`, `AppToast`, `AppProgressBar`, `PinDots`, `Keypad`, `LangBuilder`.

3. Architecture & folder structure

```
lib/
├─ main.dart           app entry, Hive init, Provider, TestModeOverlay
├─ routes.dart         AppRoutes.push / replace / resetTo
├─ content/           Article, MythQuestion, DailyTip, ContentRepository,
│                     BundledRepository (all offline data, EN + SI)
├─ l10n/              Lang enum, strings.dart t('key', lang: ...)
├─ models/            SavedArticle (Hive typeId 1), CycleEntry (typeId 0)
├─ screens/           all screens + stubs/ (Members 1 & 4 placeholders)
├─ services/          pin_service, session_state, saved_articles_store,
│                     metrics_service, local_storage_service
├─ theme/app_theme.dart colour tokens and ThemeData
└─ widgets/           reusable UI components
test/                 unit, widget and user-flow tests (105 tests)
docs/                 integration.md, user-testing.md, manual-testing-guide.md
```

State: `provider` supplies a single `SessionState` (in-memory unlock flag + persisted lockout). Language is read through `LangBuilder`. Content access goes through the `ContentRepository` interface so a remote source could replace `BundledRepository` later.

4. Navigation & app flow

`MainScreen` hosts a bottom navigation bar with three tabs: **Dashboard** (Member 2), **Calendar** (Member 2) and **Learn** (Member 3, the Education Hub).

```
PrivacyLockScreen --(correct PIN)--> MainScreen
MainScreen → Learn tab → EducationHubScreen
  └─ Our Body / Food & Diet → CategoryScreen → ArticleScreen
  └─ Daily Tip "Read More"  → ArticleScreen
  └─ Search results         → ArticleScreen
  └─ Saved Library          → OfflineLibraryScreen → ArticleScreen
  └─ Myth Buster            → MythBusterScreen
HIDE (any screen) → MainScreen (stack cleared, no animation)
```

Current setting: at your request, `main.dart` opens `MainScreen` directly (the PIN screen will be wired in later). If `PrivacyLockScreen` is ever reached with no PIN set, it unlocks the session and forwards to `MainScreen`; the PIN-setup stub has a "Continue to App" button. To test the lock, restore `home: PrivacyLockScreen()` and run with `--dart-define=DEMO_PIN=1234`.

`AppRoutes` centralises navigation: `push`, `replace`, and `resetTo` (clears the whole stack with a zero-duration `PageRouteBuilder`; used by HIDE).

5. Privacy features

PIN lock (REQ-3.1)

- PIN hashed with **PBKDF2-SHA-256**, random 16-byte salt (`Random.secure()`), 10,000 iterations; stored as `{salt, hash}` – never plain text.
- Unlock state lives **in memory only**; every cold start requires the PIN.
- 5 failures → 30-second lockout with visible countdown; stored in Hive so an app restart does not reset it.
- Android Back is blocked on the lock screen (`PopScope(canPop:false)`).
- A 4-digit PIN deters family members; it is not strong security – documented honestly in code comments.
- Dev seed: `--dart-define=DEMO_PIN=1234` creates a PIN on first run.

Panic HIDE button

- High-contrast pill with icon + label, top-right of the article (and other screens). One tap, no confirmation.
- Dismisses toasts, locks the session, and calls `AppRoutes.resetTo(MainScreen)` – the article cannot be reached with Back.
- Target < 2 s; measured in practice in milliseconds (no transition animation).

Other privacy measures

- No analytics, no third-party requests, no PII in logs.
- Fonts, content and images are bundled; `GoogleFonts.config.allowRuntimeFetching = false`.

6. Hive local storage explained

Hive is a fast, offline NoSQL key-value store. On mobile/desktop it writes `.hive` files to the app sandbox; on web it uses IndexedDB.

Box	Type ID	Owner	Contents
<code>cycle_entries</code>	0 (<code>CycleEntry</code>)	Member 2	Daily flow/symptoms/mood keyed by date
<code>saved_articles</code>	1 (<code>SavedArticle</code>)	Member 3	<code>articleId</code> , <code>savedAt</code> ; key = <code>articleId</code>
<code>settings</code>	– (dynamic)	Shared	keys: <code>pin</code> , <code>pin_lock</code> , <code>lang</code> , <code>profile</code> , <code>quiz</code>

Initialisation in `main()` : `LocalStorageService.init()` (`Hive.initFlutter()` + `CycleEntry` adapter), `SavedArticlesStore.init()` (`SavedArticle` adapter + box), `MockPinService.init()` (settings box).

Access pattern: widgets never call raw `Hive.box` ; they use `SavedArticlesStore` (`save` , `remove` , `toggle` , `isSaved` , `all` , `count`). `SavedArticlesStore.listenable` (`box.listenable()`) drives a

`ValueListenableBuilder` , so the hub's "{n} Guides Offline" count and the library list update instantly after a save or removal.

settings key	Owner	Value
pin	Member 1 / mock	{salt, hash}
pin_lock	Member 3	{fails, lockedUntil}
quiz	Member 3	{bestScore, lastPlayed}
lang , profile	Member 1	'en'/'si'; optional displayName

Hive type IDs are permanent. Member 3 models start at `typeId: 1` . After changing a `@HiveType` , run `dart run build_runner build --delete-conflicting-outputs` .

7. Content, bilingual support & offline behaviour

- **Content types:** `Article` , `MythQuestion` , `DailyTip` , blocks (`HeadingBlock` , `ParagraphBlock` , `ListBlock` , `TipBlock`), each text field a `Localized<T>` with `en` and `si` .
- **Seed articles (7):** `iron-rich-foods` , `water-hydration` , `foods-to-limit` , `understanding-your-cycle` , `what-is-a-period` , `hygiene-basics` , `hygiene-at-school` . IDs are stable for deep-linking.
- **Myth Buster:** 5 questions, shuffled each run; daily tips (5–7) chosen deterministically from the date.
- **UI strings:** `t('key', lang: lang)` from `l10n/strings.dart` ; tests ensure English and Sinhala key sets match.
- **Offline:** zero network calls. Content from `BundledRepository` , fonts from `assets/google_fonts/` (Outfit, Noto Sans Sinhala), data in Hive.
- Sinhala text and health content are marked for native-speaker / health-professional review.

8. Design system & colour palette

The Learn pages now share the Dashboard palette (dark maroon removed from the Learn UI).

Token	Value	Use
primaryColor	#E97495	Primary buttons, accents, highlighted topic tile
Gradient end	#F493AC	Daily Health Tip card (same as Dashboard hero)
backgroundColor	#FDF2F8	Screen background (also Myth Buster)
cardColor	#FFE4E6	Chips, tiles, progress track
textDark	#4C1D95	Titles and body emphasis
textLight	#9CA3AF	Secondary text
success	#2E9E6B	Saved state, offline banner

Pill buttons (StadiumBorder), 16–24 px card radius, soft pink shadows, calm 150–250 ms motion, Outfit font (Noto Sans Sinhala for **si**).

9. Accessibility

- **Semantics** /tooltips on icon-only controls (back, HIDE, delete key); toast announced as a live region.
- Tap targets ≥ 48 dp; keypad keys ≥ 64 dp.
- Layouts verified at 1.5 $\times$ text scale without overflow; reduced-motion respected (shake skipped).
- Long Sinhala strings never clipped (line height ≥ 1.6).

10. User-testing instrumentation

Enabled with `--dart-define=TEST_MODE=true`. `MetricsService` records timestamped events in memory only (no PII, no network): `lock_shown`, `pin_key`, `pin_error`, `unlocked`, `nav`, `article_open`, `save_offline`, `toast_shown`, `hide_tap`, `hide_done`.

Metric	Target
Taps after unlock (hub → save)	≤ 4
HIDE latency	< 2 s (typically < 100 ms)
Task success	unlocked → article opened → saved → HIDE
Error rate	wrong PIN attempts
Time on task	unlock → HIDE

`TestModeOverlay` shows a draggable **TEST • n taps** badge. Tap it for a dashboard with metrics, **Reset Session** and **Export JSON** (copies to clipboard). The Think-Aloud script (5 participants), facilitator text and observation sheet are in `docs/user-testing.md`.

Core task: unlock → Food & Diet → Iron-Rich Foods → Save → HIDE.

11. Automated tests

105 tests pass; `flutter analyze` reports only pre-existing issues in other members' files.

Area	Examples
Unit	PIN hash/verify, lockout, SavedArticlesStore, quiz scoring, content validation (EN+SI, ids, categories), ARB key parity, MetricsService
Widget	Keypad, AppToast, AppChip, HideButton, TestModeOverlay, each screen
User flows	Hub → Food → Article → Save in 3 taps; Library flow; Quiz flow with HIDE
Accessibility/offline	Sinhala rendering on all screens, 1.5× text scale, reduced motion, no network

Run: `flutter test` and `flutter analyze`.

Testing tip: Hive file I/O inside widget tests must be wrapped in `tester.runAsync(...)` so it resolves on the real event loop rather than hanging `FakeAsync`.

12. How to run the app

Goal	Command
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Normal run	<code>flutter run</code>
With metrics overlay	<code>flutter run --dart-define=TEST_MODE=true</code>
With a demo PIN (lock screen)	<code>flutter run --dart-define=DEMO_PIN=1234</code>
Web (Edge/Chrome)	choose the browser device when prompted
Tests	<code>flutter test</code>

13. Manual UI test walkthrough

1. **Hub:** tap the *Learn* tab. Check greeting, search, Daily Tip card (pink gradient), 2×2 topics.
2. **Search:** type "iron" or "cramp"; results filter instantly; open one.
3. **Category:** Food & Diet → chips All Tips / Cramp Relief / Vitamins filter the list.
4. **Article:** open Iron-Rich Foods; verify Village Tip. Tap Save → green "✓ Saved Offline" + toast; tap again to un-save.
5. **HIDE:** tap HIDE → instant return to mainScreen; Back cannot reopen the article.
6. **Library:** Saved Library tile shows the count; open the list, Read Offline, remove, view empty state.
7. **Myth Buster:** answer 5 questions, read feedback, check results and personal best, Play again.
8. **Test mode:** with `TEST_MODE=true`, drag the badge, open metrics, export JSON, reset.

A fuller checklist is in `docs/manual-testing-guide.md`.

14. Integration guide for other members

Member 1 – Onboarding & PIN

- Replace `PinSetupStubScreen`; call `pinService.set(pin)` to store the hash; then navigate onward.
- Write `settings['lang']` ('en'/'si') – Member 3's screens rebuild automatically.
- Optional `settings['profile'] = {displayName}` adds a name to the hub greeting.
- `PinService` contract: `isSet()`, `verify(pin)`, `set(pin)`.

Member 2 – Tracking

- `MainScreen` tabs: 0 Dashboard, 1 Calendar, 2 Learn.
- Deep-link to articles: `AppRoutes.push(context, ArticleScreen(articleId: 'water-hydration'))` (e.g. cramps → `water-hydration`, fatigue → `iron-rich-foods`).
- Use `HideButton` and `AppRoutes.resetTo` for instant exit.

Member 4 – Pharmacy

- Replace `PharmacyStubScreen`; articles recommending a health worker can link to it.

Full details: `docs/integration.md`.

15. Phase history & notes

Phase	Delivered
0	Dependencies, bundled fonts, routes, stubs
1	Design tokens, shared widgets, content repository
2	Privacy lock, PBKDF2 PinService, SessionState, lockout
3	Education Hub, search, Daily Tip, topics
4	Category, Article, reversible save, toast, HIDE
5	Offline Library
6	Myth Buster
7	Offline hardening, Sinhala parity, accessibility
8	TEST_MODE metrics and user-testing protocol
10	Integration hand-off
Later	Direct-to-MainScreen entry; Learn pages restyled to the Dashboard palette

Open items (Phase 9 backlog, not built)

- Optional persistent HIDE FAB; “Forgot PIN” offline reset (needs Member 1); auto-lock after 30 s in background; `FLAG_SECURE` for the recents screen.
- Health content and Sinhala translations need review by a qualified person before real use.
- Re-enable `PrivacyLockScreen` as `home` once Member 1’s PIN setup is ready.